

NMAM INSTITUTE OF TECHNOLOGY, NITTE
 (An Autonomous Institution affiliated to VTU, Belagavi)
Eighth Semester B.E. (Credit System) Degree Examinations
 May - June 2023

Duration: 3 Hours

19MA8X43 – NUMBER THEORY

Max. Marks: 100

Note: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

Unit – I		Marks	BT*	CO*	PO*
1. a)	Let a, b, c be any nonzero integers. Prove that				
	(i) If $\gcd(a, b) = d$ then $\gcd\left(\frac{a}{d}, \frac{b}{d}\right) = 1$.				
	(ii) If $a bc$ with $\gcd(a, b) = 1$ then $a c$.				
	(iii) If $a c$ & $b c$ with $\gcd(a, b) = 1$ then $ab c$.	6	L*3	1	1
b)	Use the Euclidean algorithm to obtain integers x & y satisfying $\gcd(1398, 324) = 1398x + 324y$	7	L3	1	2
c)	Solve $x \equiv 5 \pmod{11}$, $x \equiv 14 \pmod{29}$, $x \equiv 15 \pmod{31}$.	7	L3	2	2
2. a)	Prove that the congruence ' $\equiv$ ' is an equivalence relation on set of integers.	6	L2	2	1
b)	Determine all integer solutions of $405x + 141y = 12$.	7	L3	1	2
c)	A certain integer between 1 and 1200 leaves the remainders 1, 2, 6 when divided by 9, 11 and 13 respectively. What is the integer?	7	L3	2	2
3. a)	Prove that the linear congruence $ax \equiv b \pmod{n}$ has a solution if and only if $d b$.	6	L2	1	1
b)	There were 63 equal piles of apple fruit put together and 7 single fruits. They were divided evenly among 23 travelers. What is the number of fruits in each pile?	8	L3	2	2
c)	Solve $3x + 4y \equiv 5 \pmod{13}$, $2x + 5y \equiv 7 \pmod{13}$.	6	L3	2	2
Unit – II					
4. a)	Let p be an odd prime and let $\gcd(a, p) = 1$ define the Legendre symbol $\left(\frac{a}{p}\right)$. Obtain $\left(\frac{1783}{7523}\right)$.	7	L3	4	1
b)	Let the integer a have order k modulo n . Prove that $a^h \equiv 1 \pmod{n}$ if and only if $k h$. Hence deduce that if $a^i \equiv a^j \pmod{n}$ then $i \equiv j \pmod{n}$.	7	L3	4	1
c)	Find the remainder of $4447^{2018} \pmod{44}$.	6	L3	3	2
5. a)	State Wilson's theorem. Find the remainder of $99!$ is divided by 103.	7	L3	3	2
b)	Define Euler's phi function and show that for $n > 2$, $\phi(n)$ is an even integer. Also compute $\phi(17640)$.	7	L3	3	2
c)	State and prove Fermat's little theorem.	6	L2	3	1
6. a)	Let p be an odd prime and $\gcd(a, p) = 1$. Prove that a is a quadratic residue of p if and only if $a^{\frac{p-1}{2}} \equiv 1 \pmod{p}$.	7	L3	4	2
b)	Solve the quadratic congruence $5x^2 + 6x + 1 \equiv 0 \pmod{23}$.	7	L3	4	2
c)	Given that 3 is a primitive root of 43, find all positive integers less than 43 having order 6 modulo 43.	6	L3	4	2
Unit – III					
7. a)	Use the Hill cipher $C_1 \equiv 5P_1 + 2P_2 \pmod{26}$, $C_2 \equiv 3P_1 + 4P_2 \pmod{26}$ to encipher the message GIVE THEM TIME.	10	L3	5	2
b)	Encrypt the plaintext message GOLD MEDAL using the RSA algorithm with key $(n, k) = (2561, 3)$	10	L4	5	2
8. a)	The message SELL NOW is to be encrypted in the ElGamal cryptosystem and forwarded to a user with public key $(43, 3, 22)$ and private key $k=15$. If the random integer chosen for encryption is $j=23$, determine the ciphertext.	10	L3	5	2
b)	When RSA algorithm is based on the key $(n, k) = (3233, 37)$, what is the recovery exponent for the cryptosystem?	10	L4	5	2

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
